

Tut 6

Ex. 9.7 Q21

$$\sum_{n=1}^{\infty} (2+(-1)^n) \cdot (x+1)^{n-1}$$

Let $a_n = (2+(-1)^n)(x+1)^{n-1}$

$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \text{Does not exist}$ (Problem) $\uparrow$

New idea: Split into two series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} 2(x+1)^{n-1} + \sum_{n=1}^{\infty} (-1)^n (x+1)^{n-1}$$

Aim: Show $\sum 2(x+1)^{n-1}$ and $\sum (-1)^n (x+1)^{n-1}$ converge $|x+1| < 1$.

$$\sqrt[n]{|a_n|} = \sqrt[n]{2+(-1)^n} |x+1|^{1-\frac{1}{n}}$$

$$1 \leq \sqrt[n]{2+(-1)^n} \leq \sqrt[n]{3}$$

$$\therefore \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = |x+1|$$

$|x+1| < 1 \Rightarrow$ The series converges.

$|x+1| \geq 1 \Rightarrow$ The series diverges.

Idea 2: Use Root Test

$$\text{Let } b_n = 2(x+1)^{n-1}$$

$$c_n = (-1)^n (x+1)^{n-1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{2(x+1)^n}{2(x+1)^{n-1}} \right| = \lim_{n \rightarrow \infty} |x+1| = |x+1|$$

$$\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| = \dots = \lim_{n \rightarrow \infty} |x+1| = |x+1|$$

$$\lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right| < 1 \Leftrightarrow |x+1| < 1 \therefore \sum_{n=1}^{\infty} b_n \text{ conv.}$$

$$\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| < 1 \Leftrightarrow \underline{|x+1| < 1}$$

$$\therefore \sum c_n \text{ conv. for } |x+1| < 1$$

$$\sum a_n = \sum b_n + \sum c_n$$

For $|x+1| < 1$, since $\sum b_n$ and $\sum c_n$ converge,
we have $\sum a_n$ conv.

$$\lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right| > 1 \Leftrightarrow |x+1| > 1 \quad \text{WRONG!}$$

WRONG!

$$\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| > 1 \Leftrightarrow |x+1| > 1$$

$\therefore \sum b_n$ and $\sum c_n$ diverge for $|x+1| > 1$

$\therefore \sum a_n$ ~~conv.~~ div. [Cannot do that]

div. + div. $\neq$ div.

For $|x+1| > 1$,

$\rightarrow x+1 > 1$ or $x+1 < -1$

[Aim: $\sum a_n$ diverges for $|x+1| > 1$]

$$a_n = (2 + (-1)^n)(x+1)^{n-1}$$

If n is even, $a_n = (2+1)(x+1)^{n-1} = 3(x+1)^{n-1}$

If n is odd, $a_n = (x+1)^{n-1}$

Case: $x+1 > 1$

$\therefore a_n \geq (x+1)^{n-1}$ for all $n \geq 1$.

$$\lim_{n \rightarrow \infty} (x+1)^{n-1} = \infty \quad \Rightarrow \quad \lim_{n \rightarrow \infty} a_n = \infty$$

$\because |x+1| > 1$

Since $\lim_{n \rightarrow \infty} a_n \neq 0$, $\sum a_n$ diverges by n -term test.
~~for $|x+1| > 1$.~~

Case ~~$|x+1| < 1$~~ $x+1 < -1$

$$a_n = \begin{cases} 3(x+1)^{n-1} & \text{for } n \text{ even} \\ \underbrace{(x+1)^{n-1}}_{\text{positive}} & \text{for } n \text{ odd} \end{cases} \rightarrow \text{large negative number}$$

Since a_n oscillates between large positive and negative numbers, the limit does not exist.

$$\therefore \lim a_n \neq 0$$

$$\therefore \sum a_n \text{ diverges when } |x+1| > 1 \quad \left[\text{by } n\text{-term test} \right]$$

$$(a) \text{ Radius of conv.} = 1$$

Def. $|x+1| < 1 \Rightarrow \underbrace{-2}_{(1)} < x < \underbrace{0}_{(2)}$

When $x = -2 \Rightarrow \sum (2 + (-1)^n)(-1)^{n-1}$ div.

When $x = 0 \Rightarrow \sum (2 + (-1)^n)$ div. $\rightarrow$ Oscillating

Need justifications

$\therefore$ Interval of conv. is $-2 < x < 0$.

Q35

$$\sum_{n=1}^{\infty} \frac{1+2+\dots+n}{1^2+2^2+\dots+n^2} x^n$$

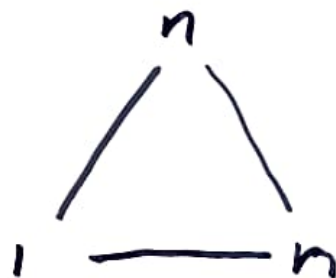
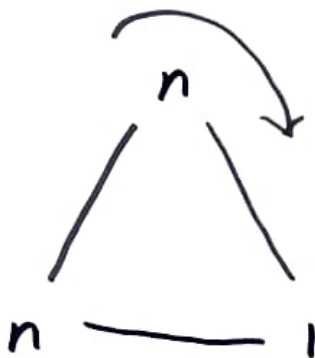
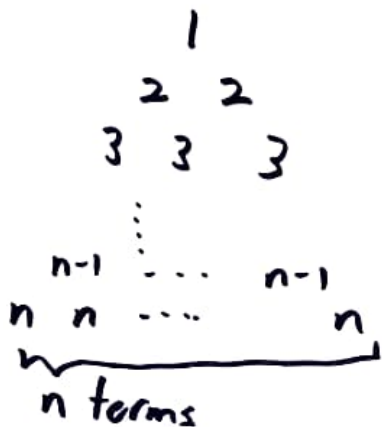
$$1+2+\dots+n = \frac{n(n+1)}{2}$$

$$1^2+2^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}$$

↓
Two 2s

↓
n ns

$3^2 \rightarrow$ Three 3s



Sum up: $1 \leftarrow 2n+1$

$2 \leftarrow 2n+1 \quad 2n+1$

$3 \leftarrow 2n+1 \quad \dots \quad 2n+1$

$\vdots$

$\ddots$

$n \leftarrow 2n+1 \quad 2n+1 \quad \dots \quad 2n+1$

$\underbrace{\hspace{10em}}$

Total number of terms

$$= \frac{n(n+1)}{2}$$

$$\therefore \text{Total values} = \frac{n(n+1)}{2} \cdot (2n+1) = 3 \left(\sum k^2 \right)$$

$$\therefore \sum k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$= 3(1^2 + 2^2 + \dots + n^2)$$

$\underbrace{\hspace{10em}}$

a triangle

$$\sum k^2$$

Ex. 9.8E

→ Quick way: $\sin x \cos x = \frac{1}{2} \sin(2x)$ Q26 $\sin x \cdot \cos x$

$$= \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \right) \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \right)$$

		$\cos x$ $\sin x$					
		x^0	x^1	x^2	x^3	x^4	x^5
x^0	0	1	0	$-\frac{1}{2}$	0	$\frac{1}{4!}$	0
x^1	1	0	1	0	$-\frac{1}{2}$	0	$\frac{1}{4!}$
x^2	0	0	0	0	0	0	0
x^3	$-\frac{1}{3!}$	0	$-\frac{1}{3!}$	0	$-\frac{1}{2}$	0	$-\frac{1}{4!}$
x^4	0	0	0	$\frac{1}{3! \cdot 2}$	0	$-\frac{1}{3! \cdot 4!}$	0
x^5	$-\frac{1}{5!}$	0	$-\frac{1}{5!}$	$-\frac{1}{5! \cdot 2}$	0	$-\frac{1}{5! \cdot 4!}$	0